

CBSE CLASS 12 CS — HIERARCHICAL CONCEPT MAP & RELATIONS

Complete Structural Visual Diagram for Solving Case-Based Networking Questions

NETWORKING & WEB SERVICES

is broadly categorized into

BRANCH 1: NETWORK PROTOCOLS

defines rules for

COMMUNICATION & TRANSFER RULES (PROTOCOLS)

TCP / IP

Core Suite

TCP: Breaks data into packets.
IP: Handles addressing & routing.

Relation: Foundation for all internet communication.

HTTP & HTTPS

Web Transfer

HTTP: Transmits web pages (plain text).
HTTPS: Encrypted web pages (SSL/TLS).

Case Key: Online banking / e-commerce → HTTPS.

specialized for

FTP

Files

File Transfer Protocol for uploading/downloading files to remote server.

Case: Transfer large files.

SMTP & POP3

Email

SMTP: Sends outgoing mail.
POP3: Retrieves incoming mail to local client.

Case: Mail transmission.

TELNET & VoIP

Remote/Voice

TELNET: Remote terminal access.
VoIP: Voice/video calls over IP networks.

Case: Video call / Remote.

connects directly via

PPP (Point-to-Point Protocol)

Data Link

Establishes a direct communication link between two network nodes (e.g., PC to ISP modem).

BRANCH 2: WEB SERVICES & INFRASTRUCTURE

provides framework for

WEB ARCHITECTURE & STANDARDS

WWW

Web System

World Wide Web. System of interlinked hypertext documents accessed via Internet.

Relation: Operates on top of TCP/IP.

HTML & XML

Languages

HTML: Web page structure & display.
XML: Custom data storage & transport.

Case: HTML displays | XML carries data.

located & accessed using

Domain Name & URL

Addressing

Domain: Human-readable host name (*cbse.nic.in*).
URL: Complete address of web resource.

Relation: DNS resolves Domain to IP.

Website & Browser

Client Side

Website: Collection of related web pages.
Browser: Client app to request & render pages.

Case: Chrome/Firefox = Browser.

hosted & served by

Web Servers & Web Hosting

Server Side

Web Server: Hardware/Software storing site assets & handling HTTP requests.
Web Hosting: Service providing online storage space on web servers 24/7.

Case Key: Host company website files so site is live continuously → Web Hosting.

⚡ INTER-CONCEPT RELATIONS & CBSE CASE STUDY SOLUTION KEYS

Secure Payment: **HTTPS** (Encrypted HTTP via SSL/TLS)

Online Meeting/Call: **VoIP** (Voice over IP)

File Uploads: **FTP** (Server file transfer)

Mail Flow: **SMTP** (Send) → **POP3** (Receive)

Data vs Display: **XML** (Carries) vs **HTML** (Displays)

Making Site Live: **Web Server** + **Web Hosting**